
Information Security

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Evolution of Information Systems

- Centralized data processing system, with a central mainframe supporting a number of directly connected terminals.
- Local area networks (LANs) interconnecting PCs and terminals to each other and the mainframe.
- Premises network, consisting of a number of LANs, interconnecting PCs, servers, and perhaps a mainframe or two.
- Enterprise-wide network, consisting of multiple, geographically distributed premises networks interconnected by a private wide area network (WAN).
- Internet connectivity, in which the various premises networks all hook into the Internet and may or may not also be connected by a private WAN.

Types of Network

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graph TD; A[Types of Network] --> B[PAN]; A --> C[LAN]; A --> D[MAN]; A --> E[WAN]; B --- B1[Personal Area Network]; C --- C1[Local Area Network]; D --- D1[Metropolitan Area Network]; E --- E1[Wide Area Network];
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PAN

Personal
Area Network

LAN

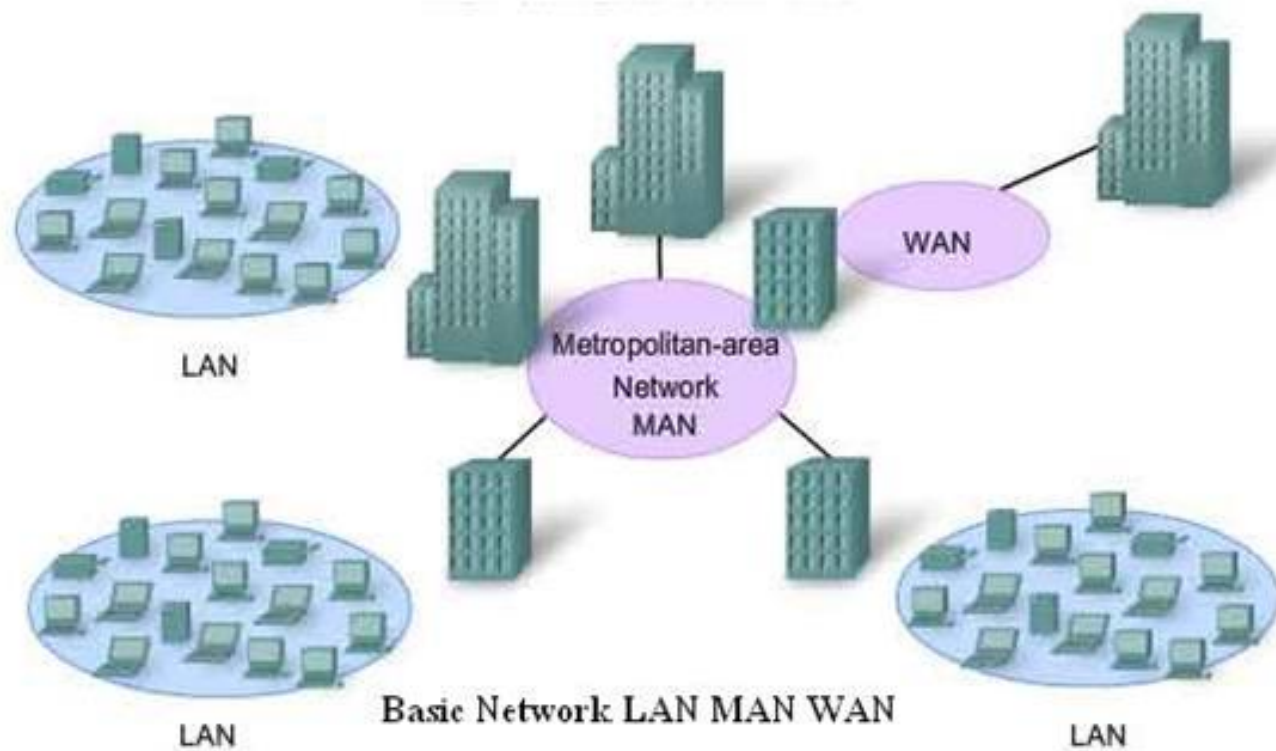
Local Area
Network

MAN

Metropolitan
Area Network

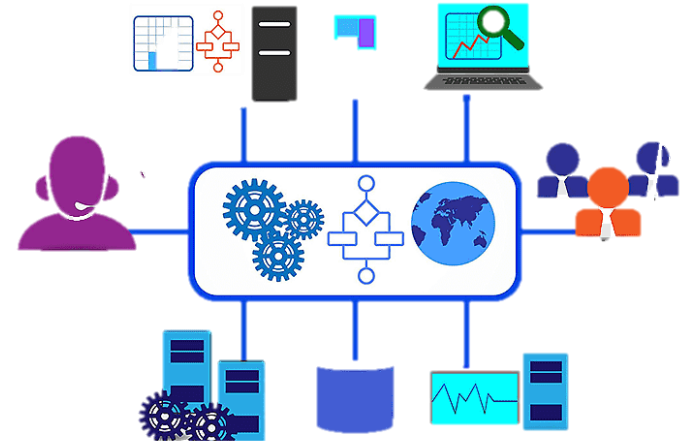
WAN

Wide Area
Network



What is Firewall?

In computing, a firewall is a network security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules. A firewall typically establishes a barrier between a trusted network and an untrusted network, such as the Internet.

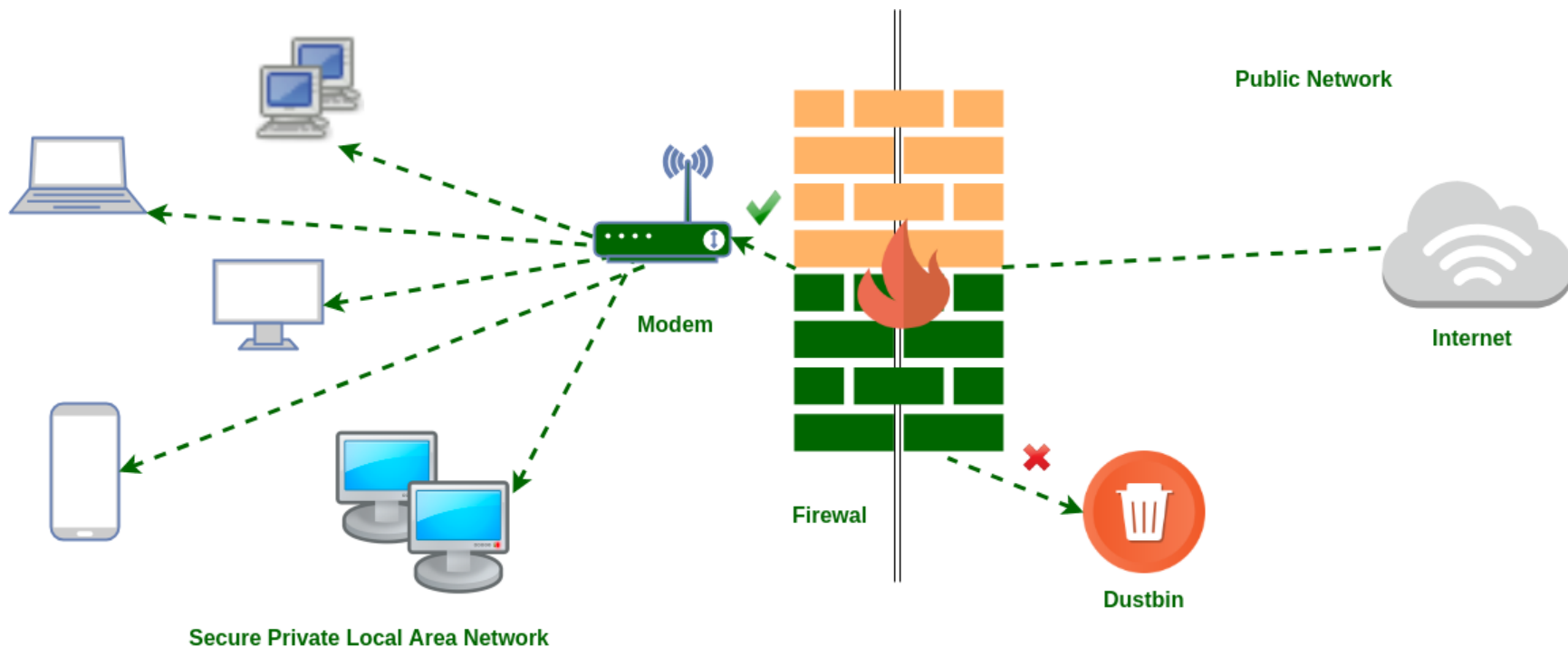


The need of firewall

- Internet access provides benefits to the organization, it enables the outside world to reach and interact with local network assets
 - a threat to the organization
- it is possible to equip each workstation and server on the premises network with strong security features
 - Not sufficient and Not cost effective
- When a security flaw is discovered, each potentially affected system must be upgraded to fix that flaw.
 - scalable configuration management
- A widely accepted alternative or at least complement to host-based security services is the firewall

Firewall Characteristics and Access Policy

- All traffic from inside to outside, and vice versa, must pass through the firewall. This is achieved by physically blocking all access to the local network except via the firewall.
- Only authorized traffic, as defined by the local security policy, will be allowed to pass. Various types of firewalls are used, which implement various types of security policies.
- The firewall itself is immune to penetration. This implies the use of a hardened system with a secured operating system. Trusted computer systems are suitable for hosting a firewall and often required in government applications.



= Specified Traffic Allowed



= Restricted Unknown Traffic

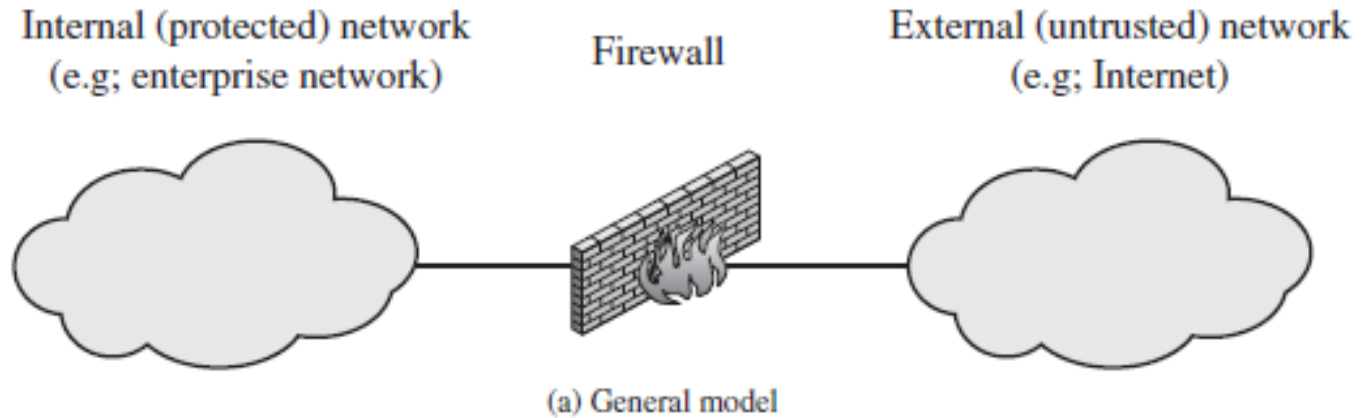
Characteristics use to filter traffic

- **IP Address and Protocol Values:** Controls access based on the source or destination addresses and port numbers, direction of flow being inbound or outbound, and other network and transport layer characteristics.
- **Application Protocol:** Controls access on the basis of authorized application protocol data. This type of filtering is used by an application-level gateway that relays and monitors the exchange of information for specific application protocols
- **User Identity:** Controls access based on the users identity, typically for inside users who identify themselves using some form of secure authentication technology, such as IPSec
- **Network Activity:** Controls access based on considerations such as the time or request, e.g., only in business hours; rate of requests, e.g., to detect scanning attempts; or other activity patterns.

Firewall Limitations

- The firewall cannot protect against attacks that bypass the firewall.
- The firewall may not protect fully against internal threats
- An improperly secured wireless LAN may be accessed from outside the organization
- A laptop, PDA, or portable storage device may be used and infected outside the corporate network and then attached and used internally.

Types of Firewalls

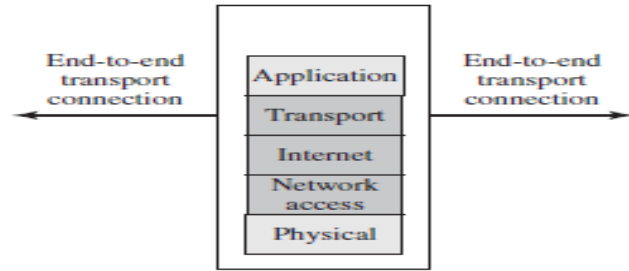


Types of Firewalls Based on Method of Operation

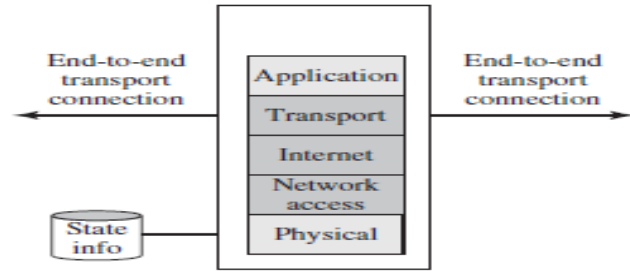
- Below is an in-depth look at the five types of firewalls based on their function and OSI layer. You can deploy any of them as hardware, software, or in the cloud.

Type of firewall	Layer(s) they operate on
Packet-filtering	Network layer (uses the transport layer to obtain port numbers)
Circuit-level gateway	Session layer
Stateful inspection	Network and transport layers
Proxy	Application layer
Next-generation	All layers except the physical layer

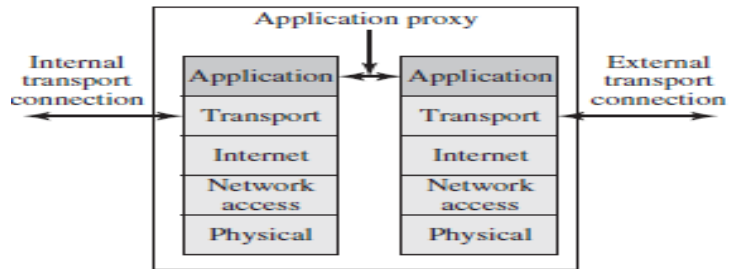
Types of Firewalls



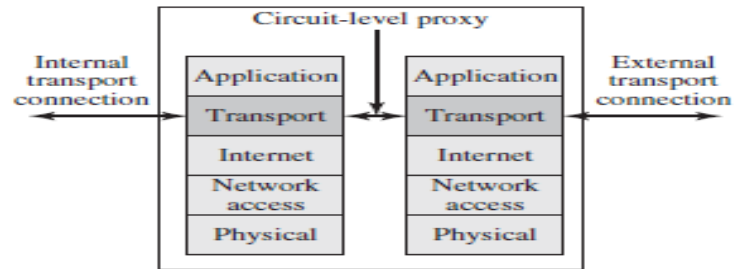
(b) Packet filtering firewall



(c) Stateful inspection firewall



(d) Application proxy firewall



(e) Circuit-level proxy firewall

Packet Filtering Firewall

- A **packet filtering firewall** applies a set of rules to each incoming and outgoing IP packet and then forwards or discards the packet
- The firewall is typically configured to filter packets going in both directions
- Filtering rules are based on information contained in a network packet:
 - **Source IP address:** The IP address of the system that originated the IP packet (e.g., 192.178.1.1).
 - **Destination IP address:** The IP address of the system the IP packet is trying to reach (e.g., 192.168.1.2).
 - **Source and destination transport-level address:** The transport-level (e.g., TCP or UDP) port number, which defines applications such as SNMP or TELNET.
 - **IP protocol field:** Defines the transport protocol.
 - **Interface:** For a firewall with three or more ports, which interface of the firewall the packet came from or which interface of the firewall the packet is destined for.

Packet Filtering Firewall

- The packet filter is typically set up as a list of rules based on matches to fields in the IP or TCP header.
- If there is a match to one of the rules, that rule is invoked to determine whether to forward or discard the packet.
- If there is no match to any rule, then a default action is taken Two default policies are possible:
 - **Default = discard:** That which is not expressly permitted is prohibited.
 - **Default = forward:** That which is not expressly prohibited is permitted

Application-level gateway

- A proxy firewall (or application-level gateway) serves as an intermediate between internal and external systems. These firewalls protect a network by masking client requests before sending them to the host.
- Proxy firewalls operate at the app layer, the highest level of the OSI model. These systems have deep packet inspection (DPI) capabilities that check both payloads and headers of incoming traffic.
- When a client sends a request to access a network, the message first goes to the proxy server. The firewall checks the following:
 - Previous communications between the client and devices behind the firewall (if any).
 - Header info.
 - The content itself.
- The proxy then masks the request and forwards the message to the web server. This process hides the client's ID. The server responds and sends the requested data to the proxy, after which the firewall passes the info to the original client.
- Proxy firewalls are the go-to option for businesses trying to secure a web application from malicious users. These systems are also popular when a use case requires network anonymity.

Circuit-level gateway

- Circuit-level gateways operate at the session OSI layer and monitor TCP (Transmission Control Protocol) handshakes between local and remote hosts.
- This simplistic firewall type quickly approves or denies traffic without consuming a lot of resources. However, these systems do not inspect packets, so even malware-infected requests get access if there's a proper TCP handshake.
- Pros of circuit-level gateways:
 - Only process requested transactions and reject all other traffic.
 - Simple to set up and manage.
 - Resource and cost-efficient.
 - Strong protection against address exposure.
 - Minimum impact on end-user experience.

Stateful Inspection Firewall

- A stateful inspection firewall (or dynamic packet-filtering firewall) monitors incoming and outgoing packets at the network and transport layers. This firewall type combines packet inspection and TCP handshake verification.
- Stateful inspection firewalls maintain a table database that tracks all open connections and enables the system to check existing traffic streams. This database stores all key packet-related info, including:
 - The source IP.
 - Source port.
 - Destination IP.
 - Destination port for each connection.
- When a new packet arrives, the firewall checks the table of valid connections. Familiar packets go through without further analysis, while the firewall evaluates non-matching traffic according to the pre-set ruleset.

Next Generation Firewalls

- A next-generation firewall (NGFW) is a security device or program that combines several functions of other firewalls. Such a system offers:
 - Deep packet inspection that analyzes the traffic's content.
 - TCP handshake checks.
 - Surface-level packet inspection.
- Next-gen firewalls also include additional network security measures, such as:
 - IDSes and IPSes.
 - Malware scanning and filtering.
 - Advanced threat intelligence (pattern matching, protocol-based detections, reputation-based malware detection, anomaly-based detections, etc.)
 - Antivirus programs.
 - Network Address Translation (NAT).
 - Quality of service (QoS) features.
 - Secure Shell (SSH) inspection.
- NGFWs are a common choice in heavily regulated industries, such as healthcare or finance. Companies that must adhere to HIPAA and PCI are the usual adopters.

Understand What Different Types of Firewalls Offer

- A firewall is the first line of defense if someone or something tries to breach your company. These systems have the potential to make or break a security strategy, so treat their selection and setup accordingly. Know what different types of firewalls offer and how they keep assets safe before you go all-in on a solution.

Thank you!